

Genomic analysis from specimen to clinical decision

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Scholar scholar.google.com/citations?user=iSjoINAAAAAJ

I have worked with genome, transcriptome, single-cell, and clinical multi-omics data since 2004, about seventeen years of professional work outside the doctoral period. The analysis carries through to reports and services that hospitals and the business actually use.

- Multi-platform NGS and multi-omics analysis, including scRNA-seq
- Clinical trial data analysis (Ph1-3) and regulatory frameworks (GCLP/CAP/ISO)
- End-to-end pipeline development with automation and QC standardization
- AI/LLM-driven variant interpretation and analysis workflows (PyTorch, TensorFlow, scikit-learn, LangChain)

2025-present **Cytogen** Chief Researcher, Diagnosis Division

Leads CTC diagnostics, scRNA-seq pipeline development, clinical report automation, and GCLP quality systems; first to define the diagnostic service portfolio and its pricing, with the market research behind it

2021-2024 Theragen Bio Director · Clinical Analysis Team / Bioinformatics

Led Ph1-3 oncology trial analysis, obtained new GCLP accreditation and designed its data management scheme, analysis lead for the national bio-big-data pilot; oversaw a large-scale WGS delivery pipeline run on AWS to meet fast-turnaround customer demand

2018-2021 **Macrogen** Deputy GM · Business Development / Clinical Analysis

Led K-MASTER genomic analysis, BRCA software with CE-IVD documentation, and liquid-biopsy commercialization strategy

2015-2018 **Yonsei University** Postdoctoral Researcher · Biomedical Systems Informatics

Multi-omics disease variant discovery and non-coding eQTL translational research across gastric cancer PDX, canine mammary tumors, pediatric allergy, and neurological disease

2012-2015 **SAIHST, Sungkyunkwan University** Postdoctoral Researcher · Molecular Informatics and Genomics

Built the miRTCat database; NRF Research Fellow studying disease variants at non-canonical microRNA target sites

2007-2012 Samsung Biomedical Research Institute Researcher · Human Genomics (2007-2011 concurrent with the Ph.D.)

Korean HapMap and KARE national cohort analysis, pharmacogenomics, and genotype imputation for the Korean population

2004-2007 Korea National Institute of Health Research Technician · Center for Genome Science

Disease-associated variant discovery in the Korean population; built and operated the SNP database

Cytogen **CTC-based liquid biopsy diagnostics**

Built and commercialized the end-to-end operating system for circulating-tumor-cell diagnostics: assay concept, pipeline, clinical report, and customer technical support.

Theragen Bio	Immuno-oncology clinical trial analysis, Ph1-3 Led oncology trial analysis for three pharmaceutical companies.
Theragen Bio	Large-scale WGS delivery on AWS Oversaw an AWS-based cloud pipeline for high-volume, fast-turnaround WGS delivery to pharma clients.
Macrogen	K-MASTER precision oncology genomics Led genomic analysis for a national precision-oncology platform.
Macrogen	BRCA1/2 variant analysis software, CE-IVD documentation Directed development of software spanning variant calling, clinical interpretation, and automated report generation, together with CE-IVD-grade documentation and performance evaluation.
2021-2024	Quality and regulatory systems Obtained new GCLP accreditation for a clinical specimen analysis laboratory, and established recurring certification cycles: CAP proficiency testing twice a year and ISO27001 annually.

■ Built

In-house · Vision	Cell image review assistant A confidence-based triage model for CTC image review that cut the cells needing human confirmation from 1,531 to 459 (about 70% routed to automatic handling). Trained on GPU and deployed for CPU inference in-house, with specimen-level cross-validation to block data leakage.
In-house · LLM	LLM-based genomic Q&A and clinical report automation An LLM system that answers from test results and source documents (RAG), together with clinical report automation.
Personal · Validation	ADMET split audit A benchmarking study on how the train/test split changes reported performance. Compared random, scaffold, and structure-disjoint splits across 12 TDC ADMET tasks plus 3 human PK tasks, on fingerprint+LightGBM, a message-passing neural network, and a fine-tuned ChemBERTa. Three initial conclusions were retracted after adding seeds, and the correction was published openly.
Personal · LLM	VeriVar A somatic variant interpretation pipeline. Repository release in preparation.
Multi-agent research	AutoBioX A multi-agent research harness that splits data preparation, prediction, verification, evidence search, and manuscript drafting.
AI agent · Hackathon	Vital STORM An agent that grades key health-screening values on a five-level scale and routes to follow-up examinations, answering from four national health-screening manuals through retrieval (RAG).
Open source · Agent harness	econ-radar An agent harness that assembles and publishes a daily economic digest weighted toward pharma, bio, and AI.

■ Selected Publications

2020	Cross-species oncogenic signatures of breast cancer in canine mammary tumors Kim TM, ... Kim KK, ... Kim S. Nature Communications 11:3616
2019	Whole-exome and whole-transcriptome sequencing of canine mammary gland tumors First author Kim KK, Seung BJ, ... Kim S. Scientific Data 6(1):147

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| 2018 | Palmitate and minimally-modified low-density lipoprotein cooperatively promote inflammatory responses in macrophages Co-first author
Ann SJ†, Kim KK†, ... Miller YI. PLoS One 13:e0193649 |
| 2013 | miRTCat: a comprehensive map of human and mouse microRNA target sites including non-canonical nucleation bulges First author
Kim KK, Ham J, Chi SW. Bioinformatics 29(15):1898-1899 |
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The full list of 20+ papers is on Google Scholar.

■ Talks

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| 2026.07 | The AutoBioX multi-agent pipeline
PseudoCon 2026 · talk |
| 2026.01 | Automated Clinical Reporting for Oncology NGS: Dual-Mode LLM Pipeline
ABDD Summit 2026 at Stanford · i-Talk and poster |
| 2025.11 | Single-cell QC standardization and an scRNA-seq × immunofluorescence pipeline
Future of Medicine Symposium: At the Crossroads of Computational Pathology and Spatial Biology · lightning talk |
| 2024.10 | Poster presentation, ABDD Summit
Awarded Best Poster Presentation |
| 2022.04 | Cancer: a history of the emperor of all maladies
1st K-BioX Global Mentoring EXPO · invited talk |
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■ Writing

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| 2026 | Generative AI in biotech and drug discovery: technology, cases, and ecosystem outlook BRIC View 2026-T24
A trend report separating how strong the evidence is at each stage, from protein design through the clinic. |
| 2026 | Integration and expansion of bio big data: data, AI, and industry BRIC View 2026-T11 |
| 2025 | Single-cell analysis in drug discovery and patient selection BRIC View 2025-T29 |
| 2018 | Discovery trends and outlook for allergic disease therapeutics BRIC View 2018-T16 |
| 2017 | Cloud-based management and services for genomic information BRIC View 2017-T14 |
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■ Awards

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| 2025 | Cheiron ResearchThon, Creative Academic Impact Award · liquid-biopsy multi-omics model for tumor microenvironment prediction |
| 2024 | ABDD Summit, Best Poster Presentation Award · Merit Awards |
| 2021 | Medical Informatics Advanced Course, Encouragement Award · Asan Medical Center |
| 2019 | Korea Clinical Datathon, Grand Prize · sepsis prediction, KoNECT |
| 2011 | KSMCB Annual Meeting, Outstanding Poster Award |
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■ Education

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| 2021-2023 | Korea National Open University B.S. Health & Environment, Business (double major) |
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2007-2011 **Sungkyunkwan University** Ph.D. Molecular Medicine

2002-2004 **Korea University** M.S. Life Science

1996-2000 **Dongguk University** B.S. Computer Science

■ **Certifications and Languages**

- Engineer Information Processing (HRD Korea, 2006); Linux Master Level 2 (KAIT, 2006)
 - English: OPIc IM2. Technical writing and business communication; one year of language study in the United States
 - Machine learning and data analysis: Coursera (Johns Hopkins Univ.), KDT AIFEL Data Scientist course (2024)
 - AI-driven drug discovery data analysis and SaMD model development (KOHI · KSA, 2025); Biohealth AX Executive Strategy Program (KOHI, Aug-Sep 2026)
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Updated September 2026